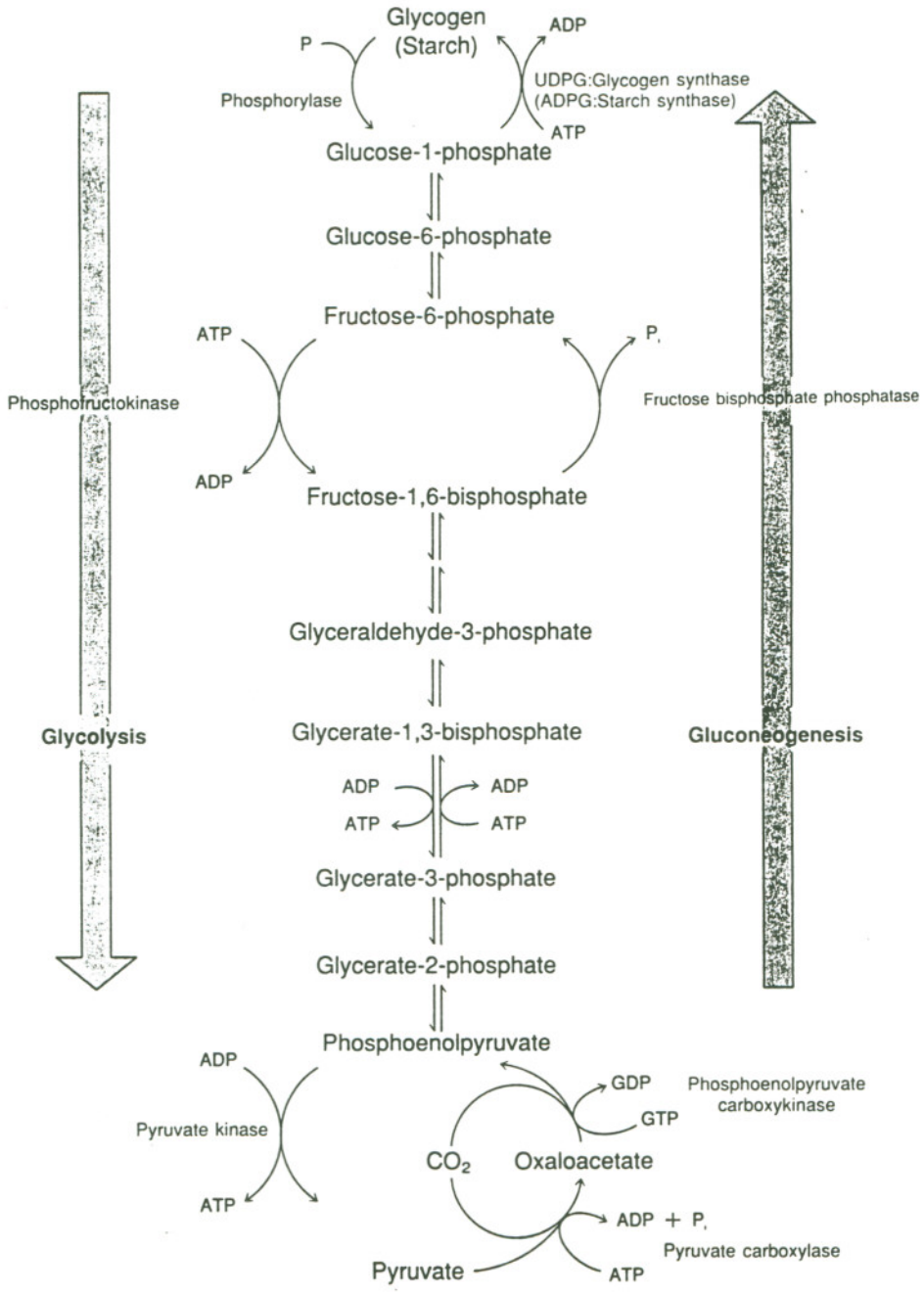
BIOCHEMISTRY CUMULATIVE EXAM

December 3, 1994

Write a complete enzymatic pathway for the synthesis of glucose from glutamate in a liver cell. You may assume that other metabolites are present in catalytic amounts, but that glutamate is the sole source of carbon and energy. Write a complete balanced equation for the reaction, including the number of ATP, CO_2 and O_2 produced or consumed. Which carbon atom(s) of glucose will <u>first</u> become labeled if glutamate labeled with C-14 at carbon number 2 is used?



$$\text{Glutamate} + NAD^+ + H_2O \longrightarrow \alpha\text{-ketoglutarate} + NH_4^+ + NADH$$

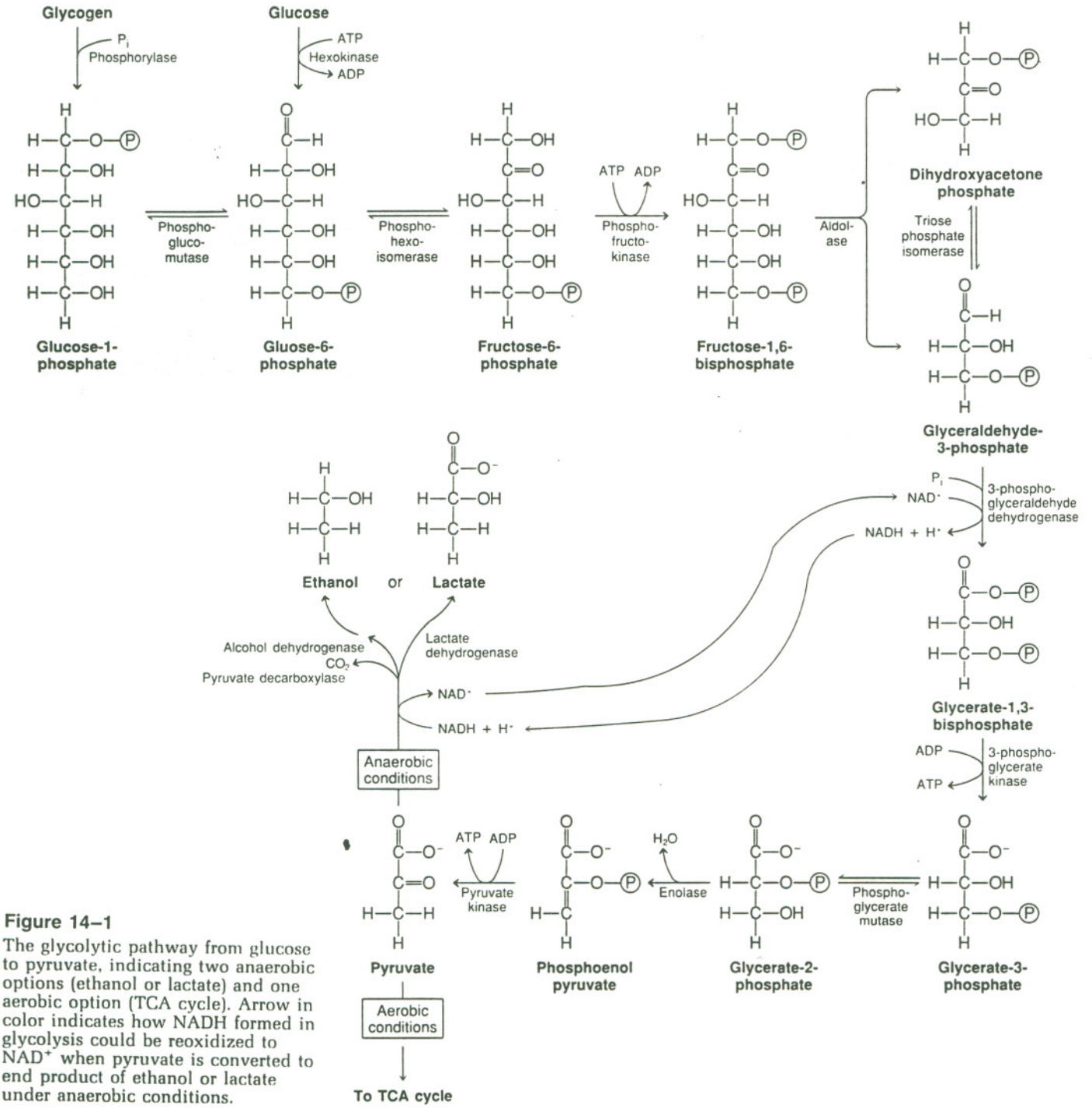
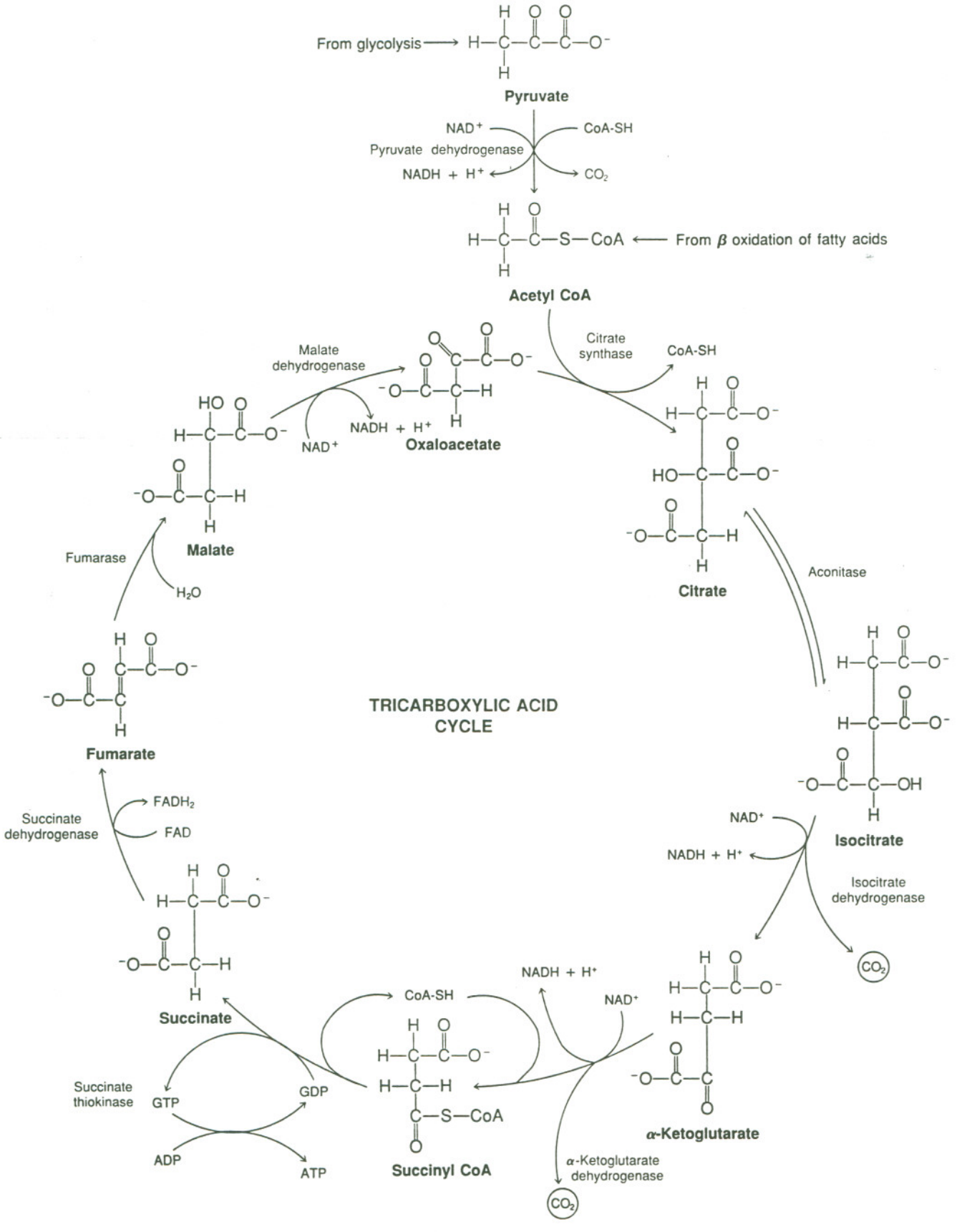


Figure 14–1
The glycolytic pathway from glucose to pyruvate, indicating two anaerobic options (ethanol or lactate) and one aerobic option (TCA cycle). Arrow in color indicates how NADH formed in glycolysis could be reoxidized to NAD^+ when pyruvate is converted to end product of ethanol or lactate under anaerobic conditions.


From glycolysis
Pyruvate
NAD+
CoA-SH
Pyruvate dehydrogenase
NADH + H+
CO2
Acetyl CoA
From β oxidation of fatty acids
Citrate synthase
CoA-SH
Citrate
Aconitase
Isocitrate
NAD+
NADH + H+
Isocitrate dehydrogenase
CO2
α-Ketoglutarate
NADH + H+
NAD+
α-Ketoglutarate dehydrogenase
CO2
Succinyl CoA
CoA-SH
Succinate
Succinate thiokinase
GTP
GDP
ADP
ATP
Succinate dehydrogenase
FADH2
FAD
Fumarate
Fumarase
H2O
Malate
Malate dehydrogenase
NAD+
NADH + H+
Oxaloacetate
TRICARBOXYLIC ACID CYCLE